

Forestation + Agriculture over Amazonia Region Dataset Analysis

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Purpose

Focus: Deforestation in the Amazonia region

Goal: Identify which factor contributes the most to changes in deforestation

Data coverage:

- Deforestation (Response Variable)
- Agriculture
- Mining
- Fire

All variables are on the same scale: land percentage of Amazonia region

Purpose:

- Determine the primary driver of deforestation change

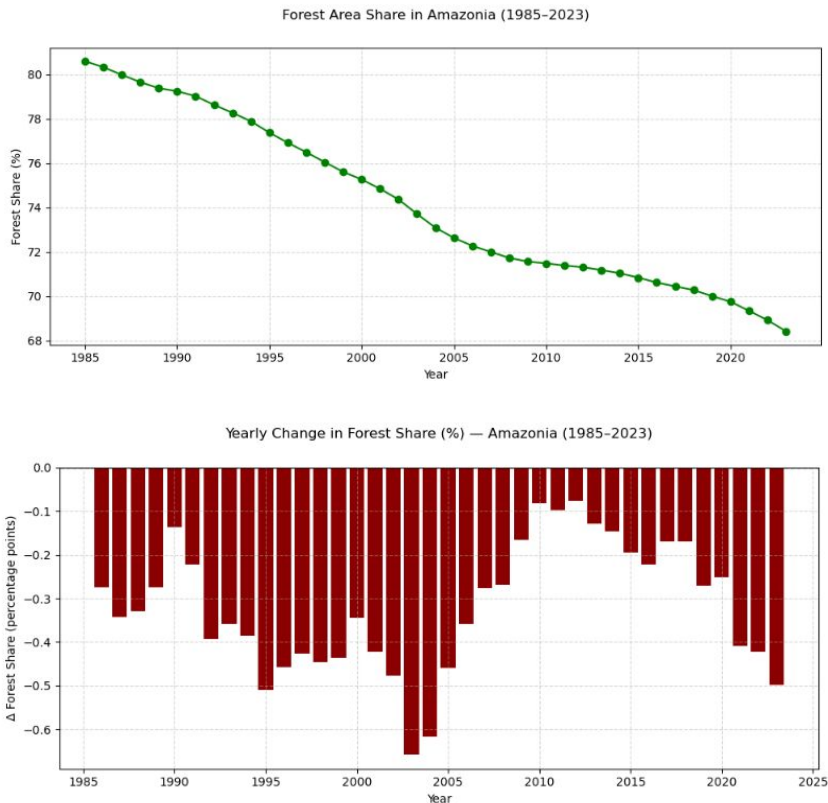
Data Description

- # of municipalities in Amazonia: 559
- Response variable: Forestation
- Feature: pasture, crop index(cassava, coffee, common bean, corn, cotton, soybean, sugarcane, wheat), mining, fire
- Timeframe: 2003-2023



Forestation

- From our forest data, the year **2003** shows the **most significant decrease in forest cover** across the Amazon region. However, based on the other datasets available, we were **unable to identify the underlying cause** of this sharp decline.
- Additionally, **another wave of deforestation began to intensify after 2012**, suggesting a renewed increase in forest loss during the following years.

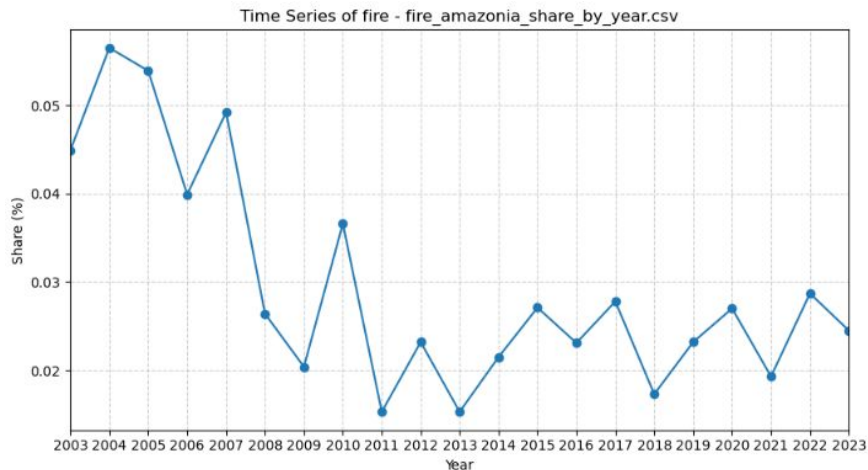


Fire

From 2003 to 2023, the share of land affected by fire in Amazonia remained **below 0.06%**, showing a generally **low but variable trend** over time.

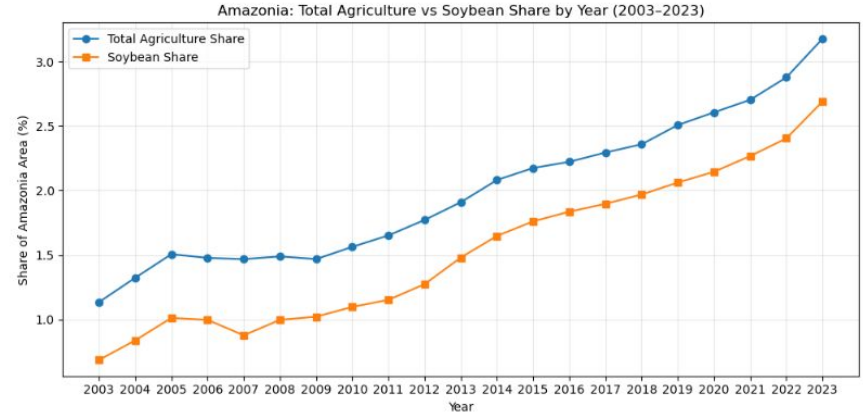
- **2004–2005** show the **highest fire activity**, reaching around **0.055%**, which may have contributed to the sharp forest loss observed in the early 2000s.
- After **2006**, fire activity **declined sharply**, reaching its **lowest point around 2010–2011 ($\approx 0.015\%$)**.
- Since **2012**, the fire share has remained **relatively stable**, fluctuating modestly between **0.015% and 0.03%**, suggesting smaller but recurring fire incidents.

Overall, the data indicate that **large-scale fires have decreased substantially since the mid-2000s**, though **small seasonal fires** continue to occur across the region.



Agriculture

- For our agricultural analysis, we combined data from multiple crops, including **bean, cassava, coffee, corn, cotton, soybean, sugarcane, and wheat**. Among these, **soybean stands out as the dominant crop** in the Amazonia region.
- Both **total agricultural area** and **soybean cultivation** have shown a **steady increase since 2003**, with a particularly **strong growth after 2009**, suggesting continuous expansion of agricultural activities—largely driven by soybean production.



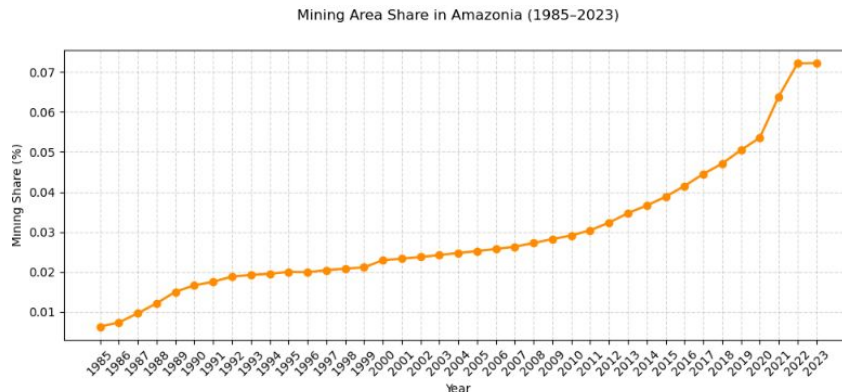
Mining

Accelerated Expansion (2010–2021):

- Starting around **2010**, the mining area began to **grow more rapidly**, reaching **over 0.06%** by **2021**.
- This indicates a **major surge in mining activities**, possibly linked to rising global demand for minerals or relaxed environmental controls.

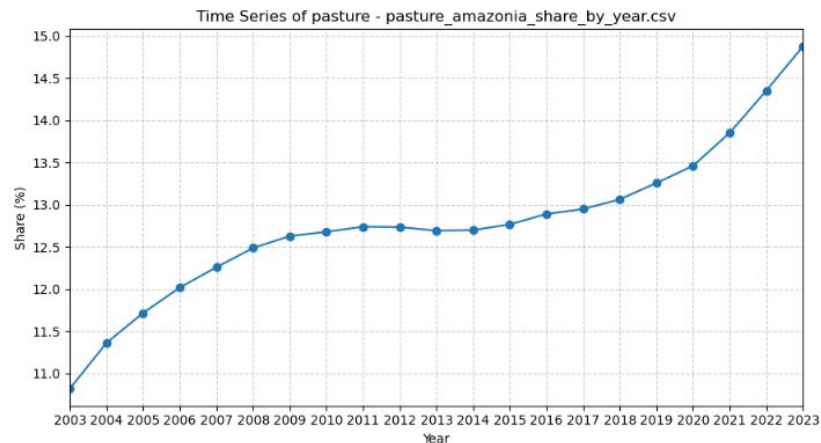
Plateau After 2021:

- After reaching its peak (~0.07%) in **2021**, the mining area **stabilized**, showing little change through 2023.
- This suggests that while expansion slowed, the overall mining footprint **remains at a historically high level**.



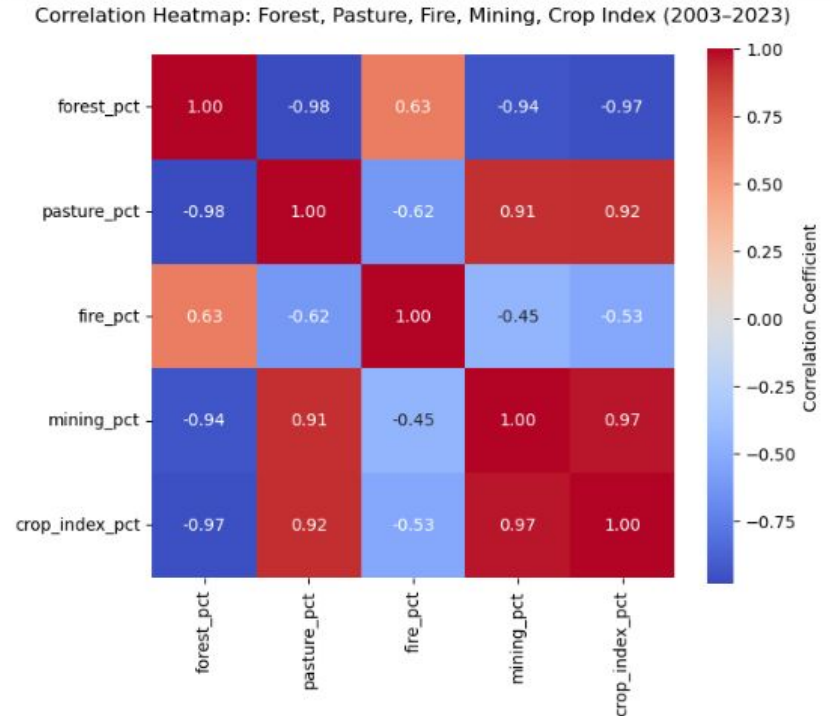
Pasture

- Pasture share in the Amazonia region **rose from ~10.9% in 2003 to ~15.0% in 2023**
- Represents a **~37% overall increase** in two decades
- The pattern shows **three distinct phases**: early expansion, mid-decade stagnation, and renewed acceleration



Correlation Analysis

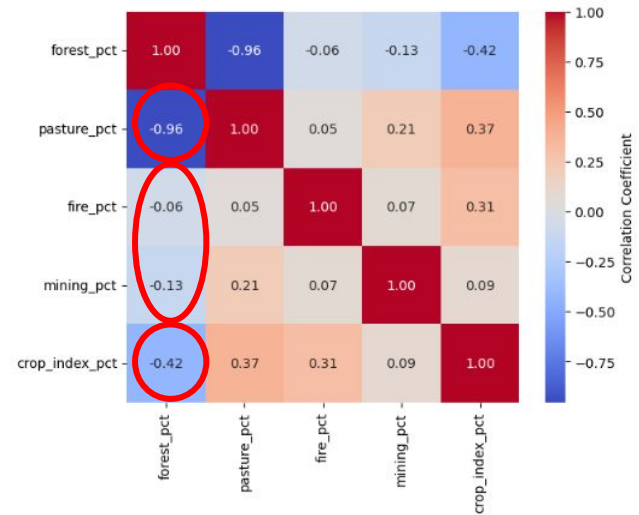
- From the correlation heatmap (2003–2023), we observe that **forest area is strongly negatively correlated** with **pasture, mining, and crop index**, indicating that expansion of these land uses directly contributes to forest loss.
- Meanwhile, **pasture, mining, and crop index** are **highly positively correlated** with each other, suggesting they often expand together across the region.
- Fire shows a **moderate correlation** with forest cover, implying that fires tend to occur **near forest boundaries** during deforestation but become **less common once land is converted** for agriculture or mining.



Correlation Analysis 2

- From the yearly-change correlation heatmap (2003–2023), we observe that **forest and pasture share the strongest negative relationship** (-0.96), indicating that when pasture area increases, forest area decreases almost proportionally.
- Changes in **crop index** also show a **moderate negative link** with forest loss, while **fire and mining** have **weak correlations**, suggesting their influence occurs over longer or less consistent periods.
- Overall, this indicates that **pasture expansion is the most immediate and consistent driver of annual deforestation**, whereas agriculture, fire, and mining contribute more gradually.

Correlation Heatmap of Yearly Changes (Δ) — Forest, Pasture, Fire, Mining, Crop Index (2003–2023)



Regression Analysis

- The regression model (**Forest_pct ~ Pasture + Fire + Mining + CropIndex**) shows an **R² of 0.99**, meaning it explains almost all variation in forest share from 2003–2023.
- Among all predictors, **pasture** and **crop index** have strong **negative effects**, confirming that **agricultural expansion is the main long-term driver of deforestation**.
- **Fire** shows a slight positive relationship, suggesting it often occurs **around remaining forest areas** rather than directly causing loss, while **mining** has **no significant effect** at this scale.
- Overall, the model highlights that **pasture and crop growth are the dominant forces behind long-term forest decline** in the Amazonia region.

$$\text{Forest_pct} = 83.2971 - 0.8192(\text{Pasture_pct}) + 7.1422(\text{Fire_pct}) + 6.1414(\text{Mining_pct}) - 9.0297(\text{CropIndex_pct}) + \varepsilon$$

===== OLS Regression Results — Forest_pct ~ Pasture_pct + Fire_pct + Mining_pct + CropIndex_pct =====

	coef	std err	t	P> t	[0.025	0.975]
const	83.2971	0.9292	89.6461	0.0000	81.4759	85.1182
pasture_pct	-0.8192	0.0816	-10.0336	0.0000	-0.9792	-0.6591
fire_pct	7.1422	2.8804	2.4796	0.0247	1.4967	12.7878
mining_pct	6.1414	7.3564	0.8348	0.4161	-8.2771	20.5599
crop_index_pct	-9.0297	1.5890	-5.6825	0.0000	-12.1442	-5.9152

R-squared: 0.994, Adj. R-squared: 0.993, n=21

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Regression Analysis 2

Our regression model ($\Delta\text{Forest} \sim \Delta\text{Pasture} + \Delta\text{Fire} + \Delta\text{Mining} + \Delta\text{CropIndex}$) shows an **R² of 0.92**, indicating a strong fit.

Among all predictors, **pasture expansion** is the **only significant factor** ($p < 0.001$), with a **strong negative effect** on forest change.

This means that when **pasture area increases**, **forest area decreases almost proportionally**, confirming pasture as the **main short-term driver of deforestation**.

Other factors — **fire, mining, and crop index** — show **no significant yearly effects**, suggesting that their impacts are more **indirect or long-term**.

$$\Delta\text{Forest} = -0.1002 - 0.7993(\Delta\text{Pasture}) - 0.0432(\Delta\text{Fire}) + 4.2761(\Delta\text{Mining}) - 1.1081(\Delta\text{CropIndex}) + \varepsilon$$

===== OLS Regression Results - $\Delta\text{Forest} \sim \Delta\text{Pasture} + \Delta\text{Fire} + \Delta\text{Mining} + \Delta\text{CropIndex}$ =====

	coef	std err	t	P> t	[0.025	0.975]
const	-0.1002	0.0212	-4.7269	0.0003	-0.1418	-0.0587
pasture_pct	-0.7993	0.0665	-11.8884	0.0000	-0.9206	-0.6600
fire_pct	-0.0432	1.0628	-0.0406	0.9681	-2.1263	2.0400
mining_pct	4.2761	4.3193	0.9900	0.3379	-4.1898	12.7420
crop_index_pct	-1.1081	1.2262	-0.9037	0.3804	-3.5115	1.2952

R-squared: 0.923, Adj. R-squared: 0.902, n=20

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Conclusion

- From **2003–2023**, deforestation in Amazonia was **mainly driven by human activity**, not natural disasters like fire.
- **Pasture expansion** had the **strongest negative effect** on forest area, confirming it as the main driver of deforestation.
- **Agricultural growth** (especially crops such as soybean) also contributed but on a smaller scale.
- The trend reflects **human migration and population growth**, leading to greater land conversion for farming.
- **Fire and mining** played **secondary or indirect roles**, with limited long-term impact compared to pasture.

Future insight

- To deepen this analysis, future work could **examine correlations and regression coefficients across municipalities** within the Amazonia region.
- This would provide a **more detailed, spatial understanding** of how different local factors contribute to deforestation.
- Expanding the dataset to include **years before 2003** would help explain the **sharp forest decline observed in 2003**, which may be linked to **climate variability or early land-use policy changes**.
- Incorporating **climate and demographic data** (e.g., temperature, rainfall, population growth, and migration patterns) could reveal **how human expansion and climate change jointly influence deforestation**.

Data Source

<http://mapbiomas.org>

https://dataserver-coids.inpe.br/queimadas/queimadas/focos/csv/anual/EstadosBr_sat_ref/

<https://ipeagit.github.io/geobr/>

Data from Tiego